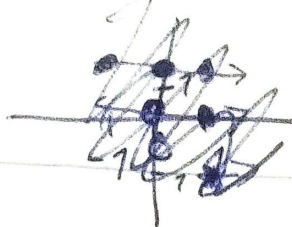


Du 4.1

A



$$\begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & -1 & 1 \\ -1 & 1 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} \approx \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 2 \\ 0 \\ 1 \end{pmatrix}$$

ti hledáme OP $\begin{pmatrix} 1 \\ 1 \\ 1 \\ 2 \\ 0 \\ 1 \end{pmatrix}$ na $\text{Im } A$
~~Koordináty~~
 Souřadnice
 v bázi $\text{Im } A$

Přes gramovu matici:

$$\left(\begin{array}{ccc|c} 4 & -1 & 2 & 2 \\ -1 & 4 & 2 & 4 \\ 2 & 2 & 6 & 6 \end{array} \right) \sim \left(\begin{array}{ccc|c} 4 & 1 & 3 & 3 \\ 0 & 5 & 5 & 7 \\ 0 & -5 & -10 & -10 \end{array} \right) \sim$$

$$\sim \left(\begin{array}{ccc|c} 1 & 1 & 3 & 3 \\ 0 & 5 & 5 & 7 \\ 0 & 0 & 5 & 3 \end{array} \right) \sim \left(\begin{array}{ccc|c} 1 & 1 & 3 & 3 \\ 0 & 5 & 0 & 4 \\ 0 & 0 & 5 & 3 \end{array} \right) \quad \begin{array}{l} c = \frac{3}{5} \\ b = \frac{4}{5} \\ a = \frac{2}{5} \end{array}$$

taková rovina je

$$Z = \frac{2}{5}x + \frac{4}{5}y + \frac{3}{5}$$

neboli:

$$2x + 4y - 5z = -3$$